

1. INTRODUCTION

Mental health refers to emotional, psychological, and social well-being. It affects how individuals think, feel, and behave in daily life. Teenagers are one of the most vulnerable groups affected by mental health problems due to academic competition, social expectations, peer pressure, and technological influence.

In today's digital world, teenagers spend a large amount of time on social media platforms. Excessive use of mobile phones and internet services often results in stress, anxiety, depression, and sleep disorders. Educational pressure and unhealthy routines further increase mental health challenges.

Data analysis techniques can help identify the major causes and trends related to mental health issues. By using Python libraries for analysis and visualization, it becomes easier to interpret data and understand the relationship between different lifestyle factors and psychological conditions.

This project performs an analytical study of teenage mental health data to identify important patterns and provide meaningful insights.

2. OBJECTIVES

The main objectives of this project are:

- To analyze teenage mental health data using Python.
- To understand the relationship between lifestyle habits and mental health.
- To perform Exploratory Data Analysis (EDA).
- To create visualizations using Matplotlib and Seaborn.
- To identify factors affecting stress, anxiety, and depression.
- To generate useful insights from the dataset.

3.Tools and Technologies Used

PROGRAMMING LANGUAGE

- Python

PYTHON LIBRARIES

- Pandas
- NumPy
- Matplotlib
- Seaborn

PLATFORM

- Jupyter Notebook
- Google Colab

4. Dataset Description

The dataset contains information related to teenagers and their mental health conditions.

Dataset Attributes

Attribute	Description
Age	Age of the student
Gender	Male/Female
Sleep_Hours	Daily study time
Sleep_Hours	Daily sleeping hours
Social_Media_Hours	Time spent on social media
Physical_Activity	Exercise hours
Stress_Level	Stress category
Anxiety_Score	Anxiety rating
Depression_Score	Depression level
Academic_Performance	Student performance
Physical_Activity	Exercise hours

5. Data Preprocessing

Data preprocessing is an important step before analysis. It improves data quality and removes inconsistencies.

✓ IMPORT LIBRARIES

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

6. Data Collection

The dataset can be downloaded from:

- Kaggle
- UCI Machine Learning Repository

Example dataset search:

- “Teen Mental Health Dataset”
- “Student Stress Analysis Dataset”
- “Student Stress Dataset”

7. Methodology

Step 1: Import Dataset

```
df = pd.read_csv("mental_health_analysis.csv")
```

Step 2: Display Dataset

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv("mental_health_analysis.csv")
print(df.head())
```

	User_ID	Age	Gender	Social_Media_Hours	Exercise_Hours	Sleep_Hours	\
0	1	16	F	9.654486	2.458001	5.198926	
1	2	17	M	9.158143	0.392095	8.866097	
2	3	15	M	5.028755	0.520119	4.943095	
3	4	17	F	7.951103	1.022630	5.262773	
4	5	17	F	1.357459	1.225462	6.196080	

	Screen_Time_Hours	Survey_Stress_Score	Wearable_Stress_Score	\
0	8.158189	3	0.288962	
1	5.151993	5	0.409446	
2	9.209325	2	0.423837	
3	9.823658	5	0.666021	
4	11.338990	5	0.928060	

	Support_System	Academic_Performance
0	Moderate	Excellent
1	Moderate	Good
2	Moderate	Poor
3	Moderate	Average
4	High	Poor

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   User_ID                               5000 non-null   int64
1   Age                                   5000 non-null   int64
2   Gender                               5000 non-null   object
3   Social_Media_Hours                   5000 non-null   float64
4   Exercise_Hours                       5000 non-null   float64
5   Sleep_Hours                          5000 non-null   float64
6   Screen_Time_Hours                    5000 non-null   float64
7   Survey_Stress_Score                  5000 non-null   int64
8   Wearable_Stress_Score                5000 non-null   float64
9   Support_System                       5000 non-null   object
10  Academic_Performance                 5000 non-null   object
dtypes: float64(5), int64(3), object(3)
memory usage: 429.8+ KB
None
```

```
print(df.describe())
```

	User_ID	Age	Social_Media_Hours	Exercise_Hours \
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	2500.500000	15.493200	4.932081	1.498151
std	1443.520003	1.715151	2.853928	0.873984
min	1.000000	13.000000	0.000528	0.000473
25%	1250.750000	14.000000	2.473150	0.734431
50%	2500.500000	16.000000	4.898176	1.483432
75%	3750.250000	17.000000	7.369195	2.276089
max	5000.000000	18.000000	9.995052	2.999774

	Sleep_Hours	Screen_Time_Hours	Survey_Stress_Score \
count	5000.000000	5000.000000	5000.000000
mean	7.057370	7.068630	3.015800
std	1.722211	2.883494	1.414762
min	4.001515	2.000481	1.000000
25%	5.611836	4.574327	2.000000
50%	7.068874	7.118979	3.000000
75%	8.519411	9.526335	4.000000
max	9.999229	11.999010	5.000000

Step 3: Check Missing Values

```
print(df.isnull().sum())
```

User_ID	0
Age	0
Gender	0
Social_Media_Hours	0
Exercise_Hours	0
Sleep_Hours	0
Screen_Time_Hours	0
Survey_Stress_Score	0
Wearable_Stress_Score	0
Support_System	0
Academic_Performance	0
dtype:	int64

Step 4: Data Cleaning

```
df.dropna(inplace=True)
```

8.Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of analyzing and understanding the dataset before applying any advanced techniques. It helps identify patterns, relationships, trends, missing values, and important insights from the data using statistical methods and visualizations.

In this project, EDA is performed on the teenage mental health dataset to study factors affecting stress levels, sleep habits, social media usage, exercise routines, and academic performance.

Objectives of EDA

- To understand the structure of the dataset.
- To identify missing and duplicate values.
- To analyze relationships among variables.
- To visualize important patterns and trends.
- To discover factors affecting teenage mental health.

9.Data Visualization Using Matplotlib-

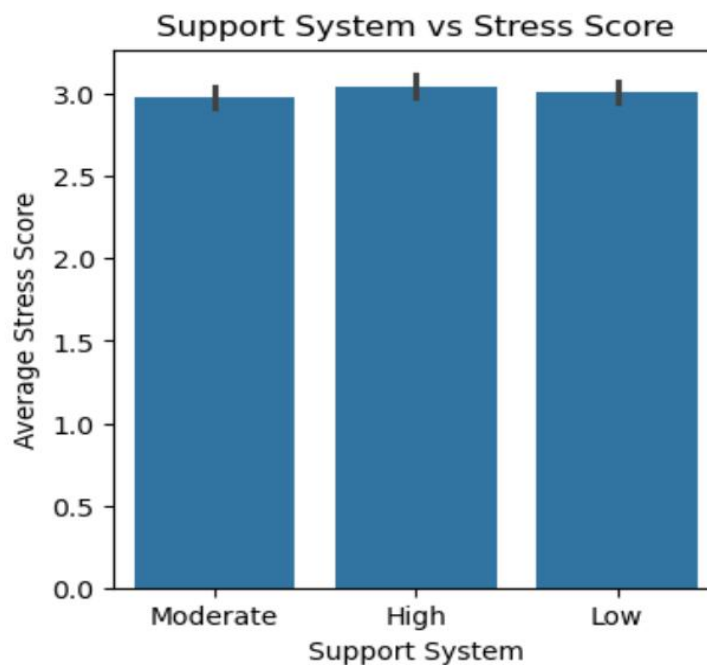
INSIGHT1: BAR GRAPH

```
plt.figure(figsize=(4,4))

sns.barplot(
    x='Support_System',
    y='Survey_Stress_Score',
    data=df
)

plt.title('Support System vs Stress Score')
plt.xlabel('Support System')
plt.ylabel('Average Stress Score')

plt.show()
```



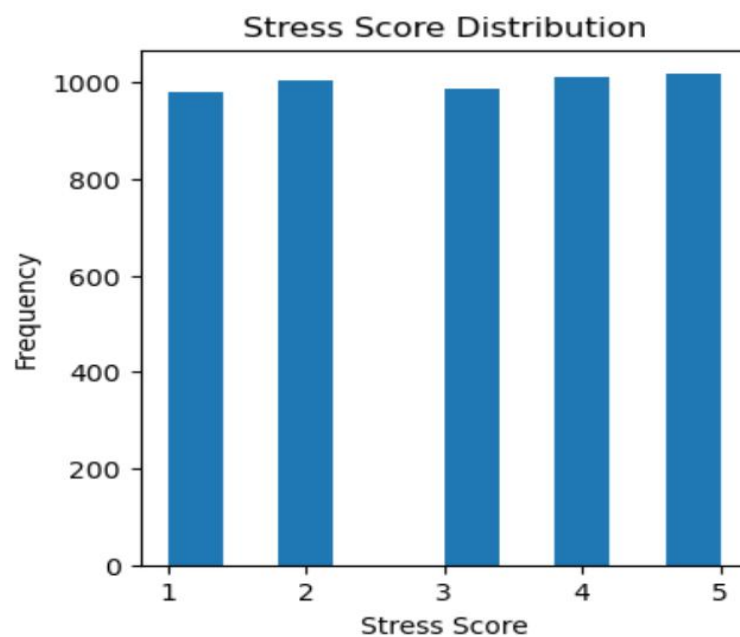
INSIGHT2: HISTOGRAM

```
plt.figure(figsize=(4,4))

plt.hist(
    df['Survey_Stress_Score'],
    bins=10
)

plt.title('Stress Score Distribution')
plt.xlabel('Stress Score')
plt.ylabel('Frequency')

plt.show()
```



INSIGHT3: PIE CHART

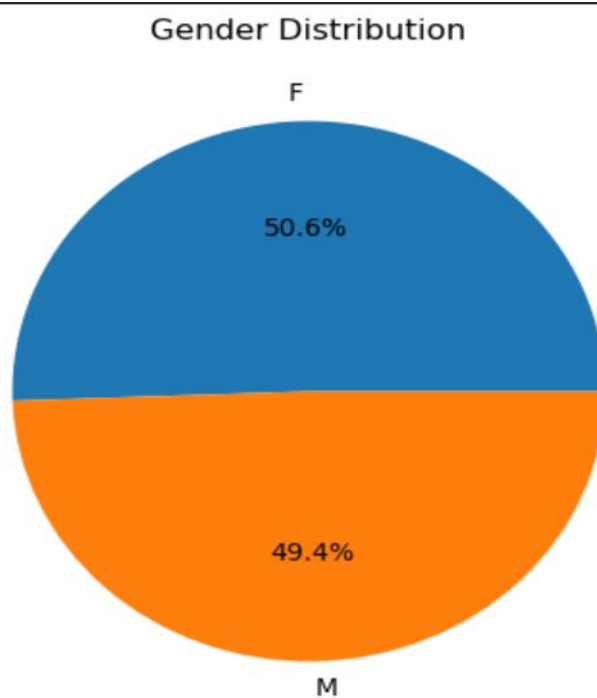
```
Gender_Count = df['Gender'].value_counts()

plt.figure(figsize=(5,5))

plt.pie(
    Gender_Count,
    labels=Gender_Count.index,
    autopct='%1.1f%%'
)

plt.title("Gender Distribution")

plt.show()
```



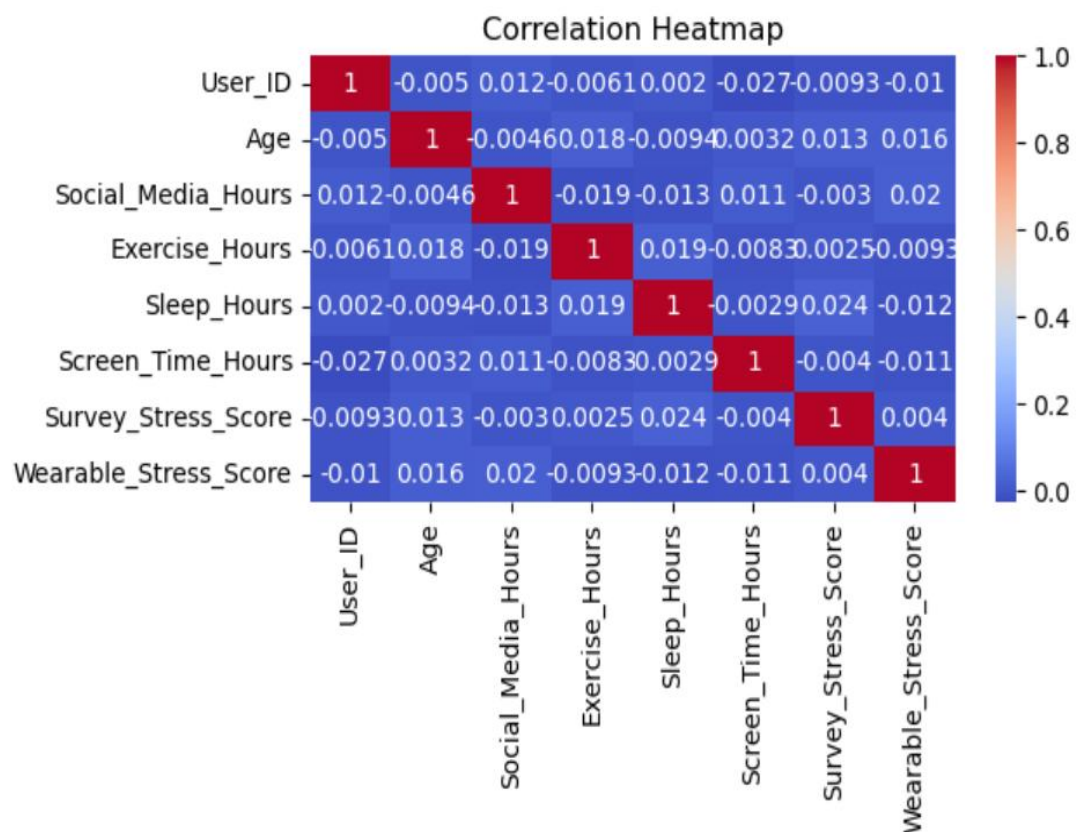
INSIGHT4: HEAT MAP

```
plt.figure(figsize=(10,6))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")

plt.show()
```



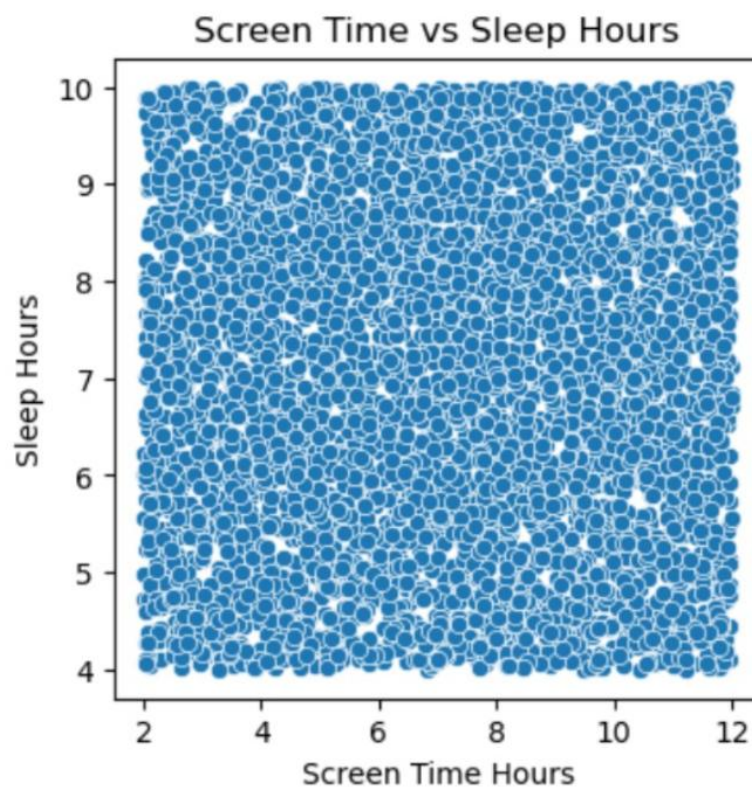
INSIGHT5: SCATTER PLOT

```
plt.figure(figsize=(4,4))

sns.scatterplot(
    x='Screen_Time_Hours',
    y='Sleep_Hours',
    data=df
)

plt.title('Screen Time vs Sleep Hours')
plt.xlabel('Screen Time Hours')
plt.ylabel('Sleep Hours')

plt.show()
```



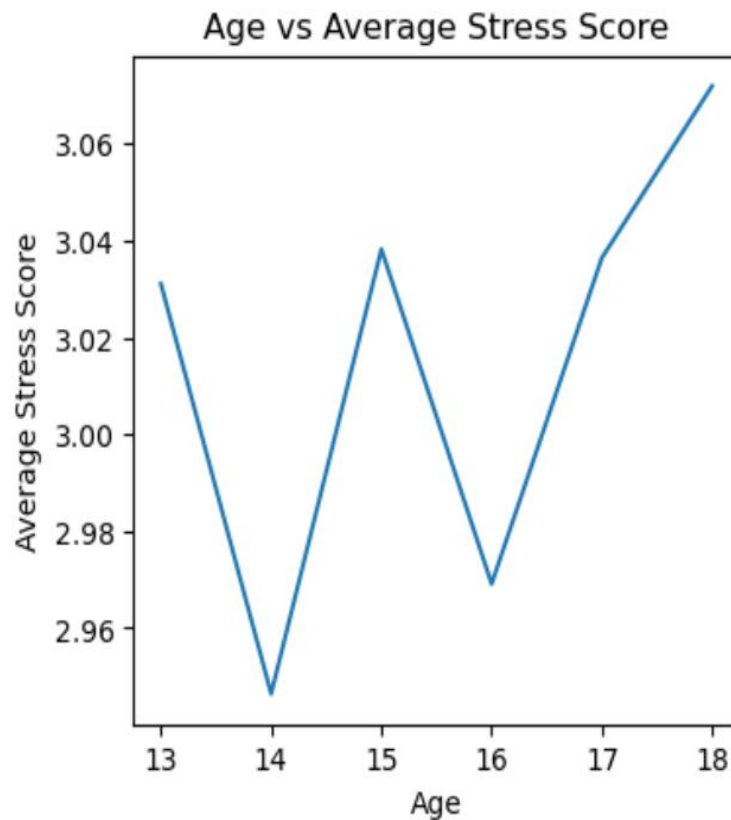
INSIGHT6: GROUP BY

```
plt.figure(figsize=(4,4))

df.groupby('Age')['Survey_Stress_Score'].mean().plot()

plt.title('Age vs Average Stress Score')
plt.xlabel('Age')
plt.ylabel('Average Stress Score')

plt.show()
```



10. Findings and Observations

- Teenagers with low sleep hours show higher stress levels.
- Excessive social media usage increases anxiety.
- Students performing physical activities experience lower depression levels.
- Academic pressure is one of the major causes of stress.
- Balanced routines improve mental health conditions.

11. Scope of the Project

The project can be useful in:

- ❖ Educational institutions
- ❖ Healthcare sector
- ❖ Student counseling systems
- ❖ Mental health awareness programs
- ❖ Research and academic studies

The analysis can help teachers, parents, and counselors understand the mental condition of student.

12. Conclusion

The project successfully analyzed teenage mental health data using Python data analysis libraries. The study revealed that sleep habits, social media usage, academic pressure, and physical activity play an important role in determining the mental health of teenagers.

The analysis showed that unhealthy routines and excessive online activity negatively impact emotional well-being, while exercise and balanced schedules improve mental health. Through visualizations and statistical analysis, meaningful insights were generated from the dataset.

This project demonstrates the importance of data analysis in understanding healthcare-related issues and highlights the need for mental health awareness among students.

13. References

- ❖ Python Official Website
- ❖ Pandas Documentation
- ❖ Matplotlib Documentation
- ❖ Seaborn Documentation
- ❖ Kaggle Datasets